Method section — What Drives an LLM's Choice? (Chris Crasto Gomes)

4. Method

This section describes the experiment exactly as run: the stimuli, the page-version history (including the mid-run change, declared in full), the trial procedure, the model set, failure handling, and cost accounting. Every figure below is taken from the code (`src/run_experiment.py` , `src/models.py`), the stimulus manifest (`pages/features.yaml`), the model configuration (`config.yaml`), the trial data (`results/results.csv` , `results/results_run3.csv`) and the cost ledger (`results/cost_ledger.md`).

4.1 Stimuli and the controlled-feature design

The stimuli are **five fictional oil-painting course landing pages** (`pages/artistry_academy.html` , `palette_masters.html` , `canvas_color.html` , `brushstroke_studio.html` , `creative_canvas.html`), served to the models as **raw HTML** (no rendering, no JavaScript). Each page instantiates a distinct offer profile drawn from a controlled feature matrix (`pages/features.yaml`), in which eleven features are varied per page: price level and display, review count, star rating, instructor credentials, course description detail, page structure, completion certificate, positioning badge, enrollment social proof, a five-way comparison table, price anchoring, and installment payment plan.

The design principle (set after two aborted pilot attempts, §4.2) is that **each page gets exactly one strength and one honest tradeoff**: star ratings are compressed into a narrow 4.3–4.8 band, review counts form a tight 180–260 ladder, only one page offers a certificate, and the five newer competitive signals (badge, enrollment social proof, comparison table, anchoring, installments) are deliberately distributed across pages rather than stacked on one. The five profiles as run:

Feature	Artistry Academy	Palette Masters	Canvas & Color	Brushstroke Studio	Creative Canvas
Price	$299 (anchored from $349)	**$149** (lowest)	$199	$249 (anchored from $299)	$179 (6 × $29.83)
Reviews	240 · 4.8★	220 · 4.6★	**260 (most) · 4.3★ (lowest)**	**180 (fewest)** · 4.5★	200 · 4.4★
Credentials	Yale MFA, MoMA exhibitor	none (studio collective)	none (community-taught, 12 yrs)	MFA Slade, former Chelsea lecturer	none (self-taught duo)
Certificate	no	no	no	**yes (only one)**	no
Lessons	96	72	64	80	**58 (fewest)**
Badge	Best Seller	Best Value	Most Reviewed	New & Trending	Editor’s Choice
Enrollment social proof	2,847 enrolled	4,120 enrolled	**6,340 (largest community)**	**410 (smallest)**	1,560 enrolled
Comparison table	yes	yes	no	yes	no
Price anchoring	yes ($349 → $299)	no	no	yes ($299 → $249)	no
Payment plan	no	3 × $49.67	no	3 × $83	6 × $29.83
Stated tradeoff	Priciest; no certificate	No certificate, no credentialed instructor	Lowest rating; no credentials/certificate/installments	Fewest reviews; second-highest price	Lowest rating; fewest lessons

All five pages are shown together in every trial — the choice is a five-way forced choice, not a sequence of pairwise comparisons. Page order within the prompt is shuffled on every trial and recorded (`page_order` column), so order effects are auditable.

4.2 Page-version history (declared)

The stimuli were not frozen from the first trial. Three page versions exist; the boundary falls inside Run 2 and is declared here rather than buried:

- **v1 — trials of 16 Oct 2026 (444 rows).** Run-2 final layout A. The controlled feature values were already at their final run-2 settings in this version: the Canvas demotion (260 reviews @ 4.3★, community-taught, no credentials) was the fix applied at the end of **pilot 2b — before any Run-2 trial**, not part of the later redesign. (An earlier draft of the outline attributed the demotion to the v2 change; that attribution was wrong and is corrected here.)
- **v2 — trials of 17 Oct 2026 (597 rows), frozen thereafter.** A **visual-layer-only** update: layouts, imagery and per-page styling changed (commit `f38306a` , *“distinct layouts per page”*); the controlled data values did not. This is verified two ways from the trial records themselves:
    1. every recorded winner attribute is byte-identical across the boundary (e.g. Artistry 240 reviews · 4.8★ · credentials on both dates; Palette 220 · 4.6★ on both);

2. the v1 models' own free-text reasoning quotes the final values — "260 reviews" appears in 52 of the 444 v1 reasonings and "4.3" in 113 — while the pre-rebalance values ("15 reviews", "4.0") appear in **zero** v1 reasonings.

- **v3 — Run 3, 22–23 Oct 2026 (210 planned / 206 completed).** v2 with exactly one change: Canvas's star rating 4.3★ → 4.6★. Everything else, and the other four pages, are byte-identical to v2 (diff-verified against `pages_run3/`).

Because models ran on both v1 and v2 inside Run 2, stimulus version is a potential confound for between-model comparisons. Which models saw which version:

Model	Versions seen in Run 2	Trials (v1 / v2)
gemini-3.1-flash-lite	**v1 only**	150 / 0
mimo-v2.6-flash	**v1 only**	150 / 0
gpt-4.1-mini	v2 only	0 / 150
grok-3-mini	v2 only	0 / 150
deepseek-flash	both	71 / 72
gpt-6-luna	both	32 / 118
gemini-3.8-flash	both	41 / 107

Two consequences are carried forward into the analysis: (i) the three dual-version models are checked for a version effect (§5.3, the DeepSeek before/after), and (ii) Run 3 is compared against Run 2 **on the same pages** — the 597 v2 trials — rather than against the pooled 1,041 (§5.5). From 17 Oct onward the pages are frozen; Run 3 is the only sanctioned change.

4.3 Procedure

One trial = one choice phase + one exit-survey phase, both from the same conversation:

1. **Choice phase.** The model receives a single prompt containing all five pages as raw HTML in shuffled order, with the instruction to *"recommend exactly ONE course that you believe is the best choice for a typical learner"*, to reason briefly over price transparency, reviews/ratings, credentials, description detail and page structure, and to end with a strict JSON line `{"choice": "page_id"}`.
2. **Exit survey.** In the **same conversation, immediately after the choice**, the model is asked why it chose that course: ten 1–5 influence ratings (price, reviews, credentials, description, clarity/structure, badge, social proof, comparison table, price anchoring, payment plan; 1 = not at all influential, 5 = very influential) plus one free-text item. Because the survey is answered post-choice with the decision already in context, it is analysed as **post-choice self-report** (§5.4), never as an independent causal measure.

Sampling parameters. Temperature is fixed at **0.1 for every provider** (stated because the study's near-determinism claims depend on it — `src/models.py` sets it per provider), with a maximum of 2,000 output tokens per call.

Retries. A trial may retry a failed or unparseable phase up to **3 attempts for the choice** and **10 for the interview**. In Run 2, 87 of 1,041 choice phases and 24 exit surveys succeeded only after retries (observed maxima: 3 and 5 respectively); in Run 3, 22 choice phases needed retries (max 3) and one survey (3). A trial that still fails after its final attempt is **dropped, not imputed** (§4.5).

Schedule. The runner iterates model × page-condition × repetition, where `condition_page` is a **scheduling/resume key only** — every trial presents all five pages regardless of the condition label (each model therefore contributes up to 30 × 5 = 150 rows in Run 2 and 30 rows in Run 3). Run 2 planned **7 models × 5 conditions × 30 = 1,050** rows; Run 3 planned **7 × 5 × 6 = 210**. The runner is resumable (completed keys are skipped on re-launch), logs one CSV row per completed trial with token counts, latencies, costs and attempts, and enforces the **$100 budget cap** at run time.

4.4 Models

Seven low-tier models across five providers were selected per supervisor direction (9 Oct 2026: *low tier only, ~2 per company*): OpenAI (×2), Google (×2), xAI, Xiaomi and DeepSeek. Anthropic was dropped on 10 Oct 2026 — the only low-tier Haiku option was a ~10× price outlier and the account was out of credit. Prices are the per-1M-token rates used for cost accounting (`config.yaml`, verified against provider pricing in October 2026):

Model	Provider	Input $/1M	Output $/1M
gpt-6-luna	OpenAI	0.10	0.50
gpt-4.1-mini	OpenAI	0.15	0.60
gemini-3.1-flash-lite	Google	0.25	1.50
gemini-3.8-flash	Google	0.75	3.75

Model	Provider	Input $/1M	Output $/1M
grok-3-mini	xAI	0.30	1.50
mimo-v2.6-flash	Xiaomi	0.14	0.28
deepseek-flash	DeepSeek	0.15	0.60 (off-peak)

All seven completed both runs; findings characterise this tier, not frontier models.

4.5 Failures, retries and dropped trials

Failure sources, from the run logs:

- **grok-3-mini** — xAI credits exhausted mid-run (150 failed attempts; all recovered after a top-up).
- **deepseek-flash** — empty/invalid responses ("No JSON with 'choice' key"): ~110 across Run 2 and the Run 3 flakes; 54 of 79 recovered by a dedicated resume pass in Run 2.
- **gpt-6-luna** — 1 credit-exhaustion event; **mimo-v2.6-flash** — rate-limit/blank retries; **gemini-3.8-flash** — 2 × "Deadline Exceeded".

All were transient and retried in-flight (§4.3). After the final retry:

- **Run 2: 9 trials dropped of 1,050 planned** — 7 deepseek-flash, 2 gemini-3.8-flash — giving **1,041 analysed (99.1% completion)**.
- **Run 3: 4 trials dropped of 210 planned** — all deepseek empty-response flakes, the same failure mode at a higher per-trial rate (a second resume pass recovered 6 of the first 10) — giving **206 analysed (98.1%)**.

Denominator policy. Percentages are computed over **completed trials** (Run 2: 143–150 per model; Run 3: 26–30 per model); dropped trials are reported as explicit counts out of the planned total and never imputed. As a sensitivity bound, even if all 7 dropped deepseek trials had been Palette picks, deepseek's Palette share would move only from 25/143 = 17.5% to 32/150 = 21.3% — no conclusion changes.

4.6 Cost accounting

Costs are computed from measured token counts × the published per-1M rates in `config.yaml`, logged per row. The authoritative figures (`results/cost_ledger.md`):

Run	Spend	Note
Run 1 (archived)	$13.59	ceiling-effect run, unusable data (interview called as a separate context)
Run 2 (incl. aborted attempts 2a/2b)	$34.16	2a $2.27 + 2b $1.24 (both aborted at design stage) are part of this total; ~$0.033 per completed row
Run 3	$6.82	206 completed trials
Total	**$54.57**	**of the $100 cap (54.6%)**

Provider billing dashboards (Google, xAI, OpenAI, DeepSeek; screenshots dated 20 Oct 2026, published with the report) are retained as receipts, but their windows are wider than the runs — the **$54.57 ledger total is the figure used everywhere in this report** (it supersedes the $50–60 estimate quoted verbally in the demo video).